Prediction Market Optimization

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Problem Statement - Why Prediction Market?

- Prediction markets are a real-money, real-time setting where beliefs update quickly.
- Categories (sports, elections, geopolitics, macro, etc.) create a natural cross-domain diversification problem.
- Core proposal idea: cap domain exposure while pursuing risk-adjusted returns with an online optimization algorithm.
- **Week 9 Changes:** Week 8 showed that a naive setup (ratio objective, heavy uniform mixing, huge universes) can look “optimized” while still behaving like equal weight. The project story is now **methodological iteration**: smoother training objective, scaled data frequency, smaller tradable set, Bayesian tuning, and honest holdout + attribution.

Process Overview

1. Ingest active events
2. Flatten + domain tagging
   - Convert event->market nested JSON to market rows, map tags to category/domain, and keep YES token IDs for binary interpretation.
3. Category-balanced market selection
   - Prioritize high-liquidity categories, apply per-category caps
4. Fetch token histories
5. Build cashed Week 8 dataset
6. Baseline run
   - Build price matrix, compute returns, allocate equal weight by domain, and save baseline metrics/time series.
7. Constrained run + running
8. Holdout evaluation + diagnostics

Core Math Formulas (1/2) - Definitions

- $r_t \in R^N$: vector of token returns at step t
- $w_t \in \Delta^N$: portfolio weights on simplex, $\sum_i w_{t,i} = 1$ and $w_{t,i} \geq 0$
- D: set of event domains/categories
- $S_d(w) = \sum_{i \in d} w_i$: total capital allocated to domain d
- L_d: domain limit for domain d

Core Math Formulas (2/2) - Concepts

$$R_t = \tilde{w}t^\top r_t, \qquad V_t = \prod s = 1^t(1 + R_s), \qquad \mathrm{DD}t = \frac{V_t}{\max s \le tV_s} - 1$$

$$\mathrm{Sortino}(R) = \frac{E[R]}{\sqrt{E[(\min(R,0))^2]} + \varepsilon}$$

$$\tilde{w}_t = (1-\alpha)\mathrm{softmax}(z_t) + \alpha$$

$$P_{\mathrm{domain}}(\tilde{w}) = \sum_{d \in \mathcal{D}} \max(0, S_d(\tilde{w}) - L_d)^2, \qquad P_{\mathrm{conc}}(\tilde{w}) = \sum_i \max(0, w_i - w_{\max})^2$$

$$P_{\mathrm{cov}}(\tilde{w}) = \tilde{w}^\top \Sigma \tilde{w} (\text{sample cov on the rolling window, with diagonal shrinkage})$$

$$B_{\mathrm{ent}}(\tilde{w}) = -\sum_i w_i \log(w_i + \varepsilon)$$

$$J_{\mathrm{md}}(R) = E[R] - \alpha_v \mathrm{Var}(R) - \beta_d E\big[\max(0, -R)^2\big]$$

Full per-step training objective (maximize; code minimizes $-$`obj`):

$$\max_{\tilde{w}} \Big[J_{\mathrm{md}}(R_{\mathrm{window}}(\tilde{w})) -_{\mathrm{domain}} (\tilde{w}) - \lambda_c P_{\mathrm{conc}}(\tilde{w}) - \lambda_\Sigma P_{\mathrm{cov}}(\tilde{w}) + \lambda_e B_{\mathrm{ent}}(\tilde{w}) \Big]$$

Evaluation Design

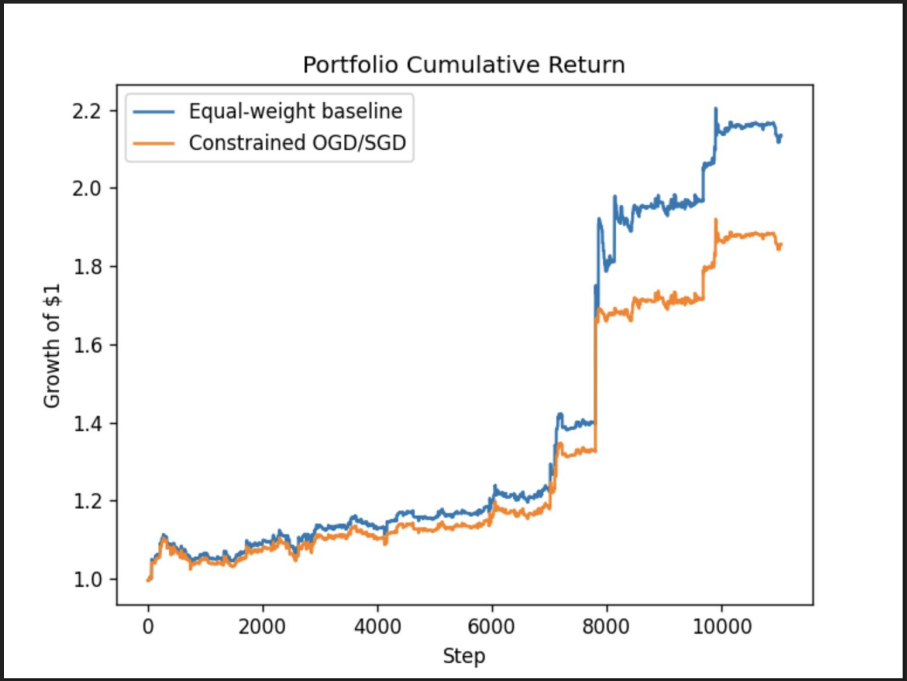
- Walk-forward tuning: Hyperparameters selected using rolling in-sample windows to maximize penalized Sortino ratio.
- Holdout testing: Final performance evaluated strictly out-of-sample on unseen data.
- Baseline comparison: Equal-weight portfolio used as benchmark for return, volatility, and drawdown.
- Metrics tracked: Cumulative return, max drawdown, volatility, rolling mean return, and return distribution

Biggest Changes

Topic	Week 8 flash guide	Week 9 (current)
Training target	Sortino ratio in-window	**Mean–downside** surrogate; Sortino for **reporting**
Uniform mix	Often high (e.g. 0.88 in old snapshot)	Search over **low** floors, including 0 (`uniform_mixes` in pipeline)
Inner optimization	Single-step narrative	`**steps_per_window=5` **, **Adam**
Evaluation	Frozen vs online was a Week 8 issue	Default **online** holdout
Universe	41-market story in old guide	Pipeline `**max_markets=40` ** (deliberate signal-to-parameter balance)
Data clock	Hourly mental model	`**history_fidelity=10**` (10-minute bars); walk-forward lengths **scaled** (~10d train / ~2d test per fold in 10-min steps)
Tuning	Grid emphasis	**Optuna** + pruner fix
Reporting	Figures only	**Automated Week 9 diagnostics** (attribution tables, correlation summary)

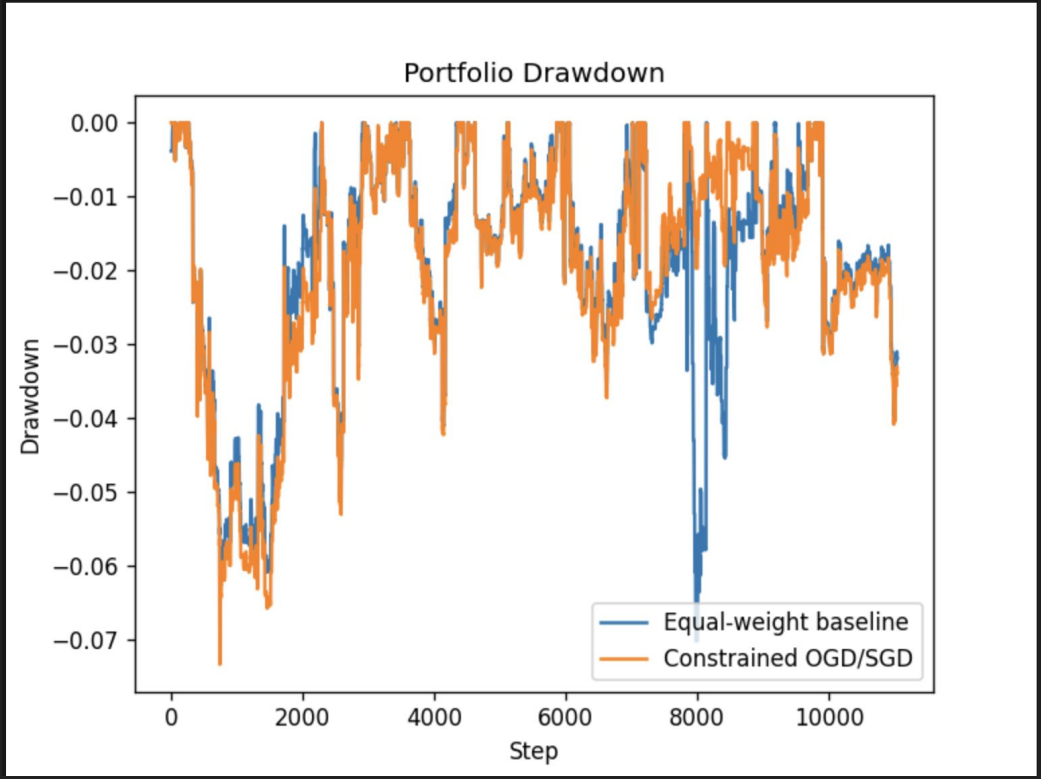
Equity Comparison

- Divergence from baseline portfolio, but still not extremely profitable



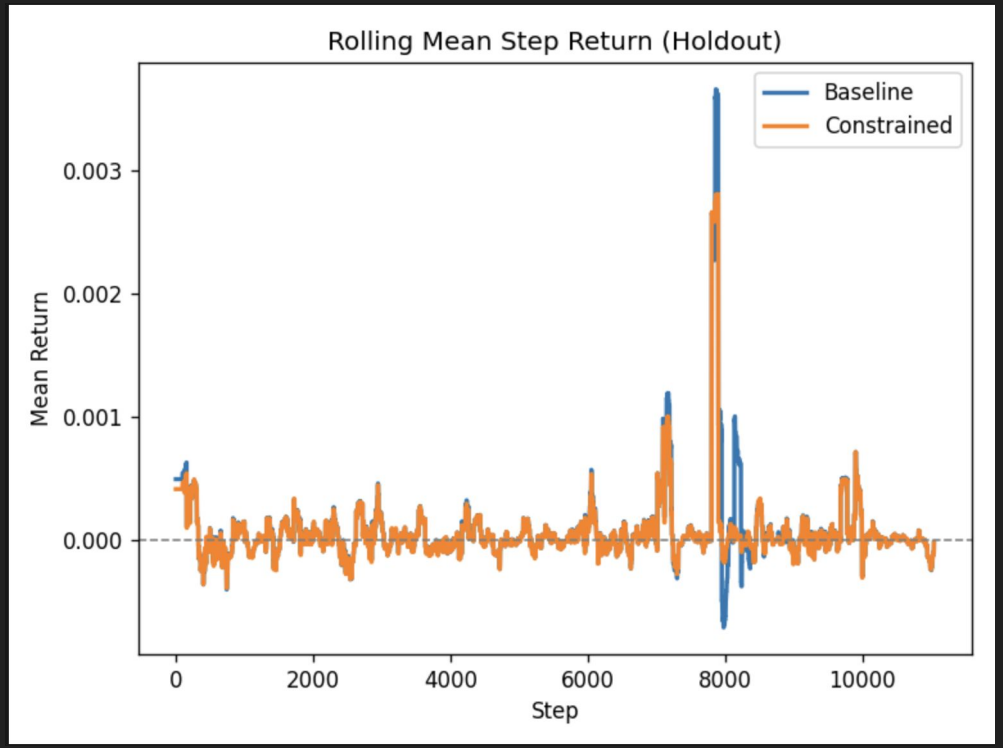
Drawdown Comparison

- Both strategies experience very similar drawdown patterns throughout the sample.
- However, our constrained portfolio avoids the largest drawdown of the baseline portfolio

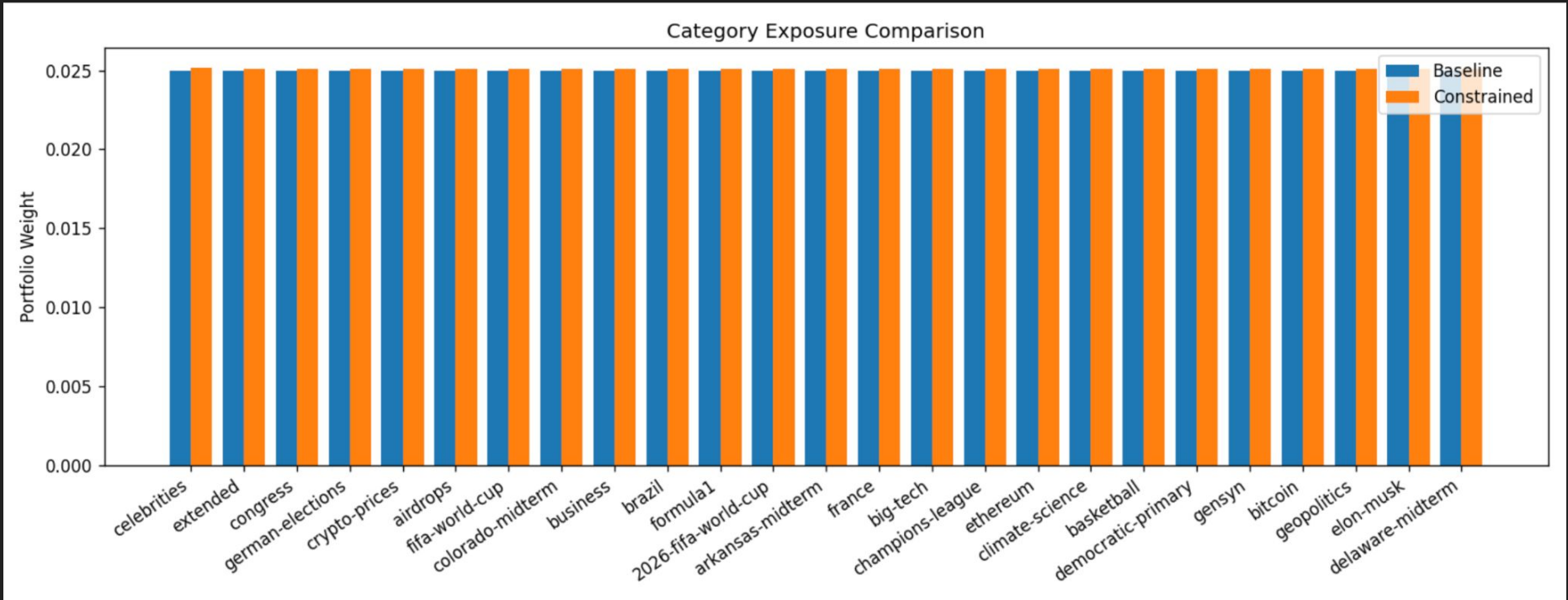


Rolling Mean Return Comparison

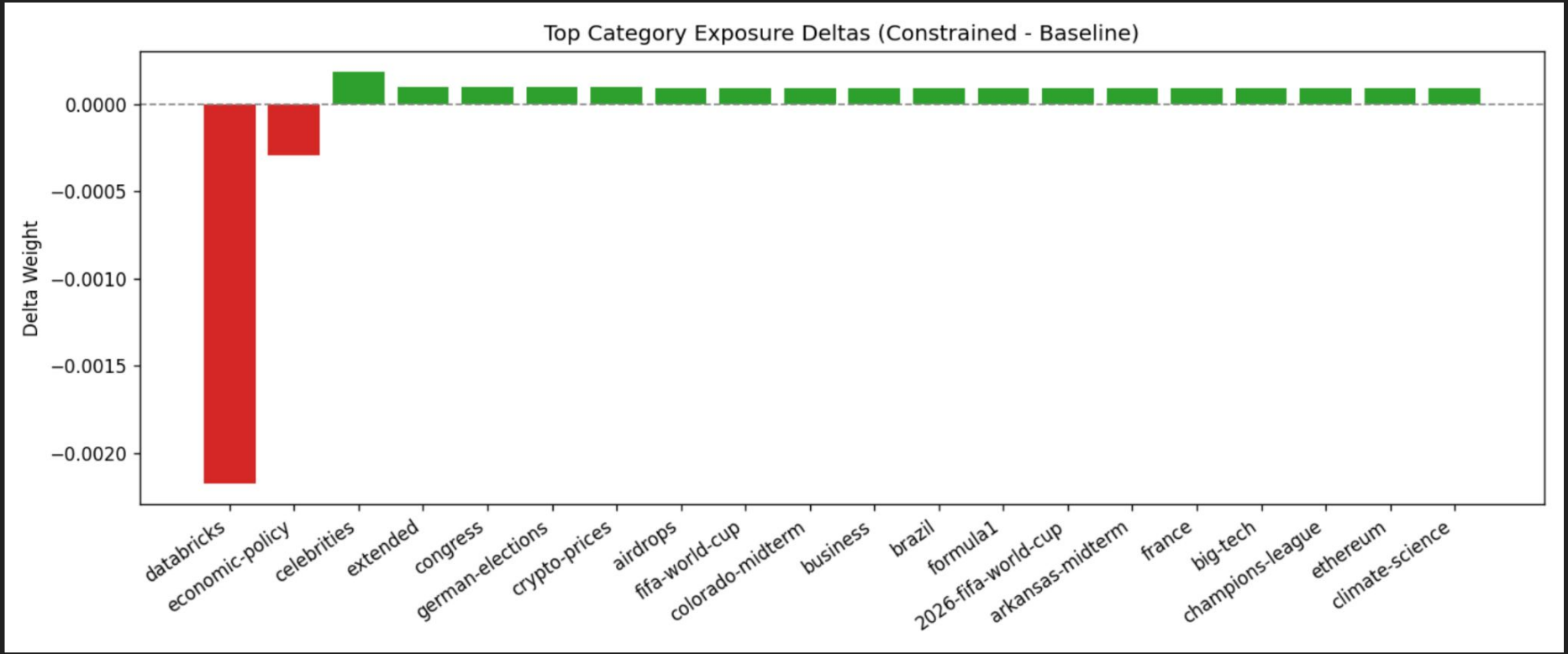
- Both strategies show highly unstable rolling mean returns that frequently cross zero.
- The constrained portfolio does not produce a more persistent positive return pattern in holdout.



Exposure Results (1/2)



Exposure Results (2/2)



Current Progress

- Markets retained: 40
- Category count: 40
- Data / eval: 10-minute bars; online holdout; walk-forward train/test block ≈ 1440 / 288 steps (~10d / ~2d per fold in 10-min steps); ~44,168 tuning steps / ~11,042 holdout steps
- Tuning: Optuna on mean–downside objective (100 trials, 80 completed in saved week8_constrained_best_metrics.json)
- Best constrained setup (selected trial): uniform_mix ≈ 0.087, domain_limit ≈ 0.13, max_weight ≈ 0.13, rolling_window = 288, evaluation_mode = online
- Holdout baseline Sortino: 0.0791 | Holdout constrained Sortino: 0.0700 (Δ −0.0091)
- Holdout baseline max drawdown: −7.16% | Holdout constrained max drawdown: −7.34%